

ALIYUS HEDRI

Freelance Robotics, Embedded Systems & IoT

Bandung, West Java, Indonesia | WhatsApp @aliyushedri | aliyus.hedri@gmail.com
linkedin.com/in/aliyushedri/ | github.com/aliyushedri | aliyushedri-portfolio.tech

Engagement: Remote freelance and project-based work. Onsite collaboration and scope are discussed per project.

PROFILE

Electrical Engineering Master by Research student at Telkom University, supporting research at ACS Laboratory. I work across robotics, SLAM, embedded systems, IoT, automation, and AI, with a practical focus on integrating sensors, firmware, software, and operator tools.

SERVICES

- **Robotics & SLAM.** Mobile and legged robot integration, LiDAR-inertial odometry / SLAM, sensor fusion, and ROS2 workflows.
- **Embedded & IoT.** Arduino, ESP32, Raspberry Pi, sensor interfacing, MQTT / HTTP, telemetry, and connected prototypes.
- **Programming.** C/C++, Python, and MATLAB for embedded control, computer vision, data work, and system integration.
- **Automation & AI.** Automation workflows, LLM integrations, Telegram bots with or without AI, and ML / DL training for image detection and processing in embedded or standalone setups.
- **Android & web.** Android Studio development for robot control and research support, plus HTML / CSS / JavaScript portfolio work on GitHub Pages.
- **Electronics & writing.** PCB design with EasyEDA / EasyEDA Pro, and technical paper or journal editing with LaTeX support.

SELECTED PROJECTS

TEL-U BOT Android application | 2026

Android Studio application developed to support TEL-U BOT robot research. The published app supports local Wi-Fi UDP control with ESP32, robot speed and direction, actuator control, line / LiDAR / odometry telemetry, calibration and settings, planner configuration, EEPROM planner read / write / test, and JSON import / export. *Published listing:* [Google Play - TEL-U BOT](#).

3D SLAM for AGV using FAST-LIO | 2026

Final-year project focused on adapting FAST-LIO for four low-cost Kinect V1 RGB-D cameras and a distributed STM32 + Mini PC architecture on ROS2 Jazzy.

Bilingual portfolio website | 2026

Designed and built a personal HTML / CSS / JavaScript portfolio for GitHub Pages, with bilingual project pages, certificate views, and downloadable CVs. [View portfolio](#).

ARJUNA hexapod robot | 2023 - 2024

Led the team developing a six-legged robot with IMU-based automatic balance and real-time YOLOv8 object detection. Team ARJUNA became a finalist in the 2024 Kontes Robot Indonesia SAR division.

EXPERIENCE & LEADERSHIP

Mechatronics Engineering Intern - PT Arthatronic Studio Teknologi | 30 Jun - 9 Aug 2025

Mechatronics engineering internship.

Lab Coordinator - EIRRG | 2024 - 2025

Electronics & Intelligence Robotics Research Group.

Basic Computer Laboratory Assistant | 2023 - 2025

Telkom University laboratory role.

Committee Member - Telkom University Student Orientation (PKKMB) | 2023

Student orientation committee.

EDUCATION

Master by Research in Electrical Engineering - Telkom University | Ongoing - Bandung

Current S2 program; research context at ACS Laboratory.

Bachelor of Engineering in Electrical Engineering - Telkom University | 2022 - 2026

GPA 3.8 / 4.0 - APERTI BUMN Scholarship Awardee 2022.

SMA Negeri Plus Provinsi Riau, Pekanbaru | 2019 - 2022

Science major - graduated with good standing.

ADDITIONAL PROJECTS

TIRNARA - IoT water-quality monitoring for fish farming | 2025

Portable biofloc water-quality monitoring product using IoT and renewable energy. Participant in PIMNAS 2025, PKM-KI class.

Household water-quality monitoring system | Feb - Jun 2025

ESP32 prototype with pH, temperature, and turbidity sensors; real-time data transmission through MQTT / HTTP to a cloud dashboard.

TOMOT - plant monitoring stick (PKM-KI) | Mar - Nov 2024

Portable soil monitoring device with pH, moisture, NPK, and temperature sensors. Bronze Medal, PIMNAS 37 presentation category.

TOOLKIT

C/C++, Python, Arduino, ESP32, Raspberry Pi, STM32, ROS / ROS2, FAST-LIO, SLAM, IMU, LiDAR, RGB-D cameras, OpenCV, YOLOv8, TensorFlow / PyTorch, NumPy / Pandas, MQTT, HTTP / REST, Android Studio, EasyEDA, EasyEDA Pro, MATLAB / Simulink, Git / GitHub, Linux, and LaTeX.

SELECTED AWARDS

- Top 20 Outstanding Students in Student Affairs - Telkom University, 2024.
- Finalist, Kontes Robot Indonesia 2024, SAR division - Team Leader, ARJUNA.
- Bronze Medal, PIMNAS 37 (2024), presentation category - PKM-KI, TOMOT.
- Participant, PIMNAS 2025 - PKM-KI class, TIRNARA.
- 2nd Place, Laboratory Competency Contest - Pekan Raya Biologi 2020, Universitas Riau.
- Participant, Madrasah Science Competition (KSM) 2021, Biology, Riau Province.

LANGUAGES

Indonesian - native. English - intermediate / working proficiency.

[Project details and supporting certificates: aliyushedri-portfolio.tech](https://aliyushedri-portfolio.tech)